

NETWORK PROBLEM 5:

DNS POISONING / DNS SPOOFING (LAYER 7 – APPLICATION LAYER)

DNS Poisoning, also known as **DNS Spoofing**, is a type of attack where a malicious actor inserts false DNS data into a DNS resolver's cache. As a result, users are redirected to fraudulent websites instead of the legitimate ones, potentially leading to phishing, malware infections, or data theft.

Causes of DNS Poisoning:

1. Compromised DNS Servers:

- Attackers can gain access to DNS servers and inject malicious records directly.
- This affects all users relying on that server for DNS resolution.

2. Man-in-the-Middle (MITM) Attacks:

- Attackers intercept DNS requests and send fake responses, redirecting users to malicious IP addresses.
- Typically occurs on unsecured networks (e.g., public Wi-Fi).

3. Software Vulnerabilities in DNS Services:

- Bugs or outdated DNS server software may allow attackers to exploit and manipulate the cache.

4. Lack of DNSSEC (DNS Security Extensions):

- Without DNSSEC, DNS responses are not digitally signed, making it easier to spoof legitimate DNS replies.

5. Cache Poisoning Attacks:

- Attackers flood a DNS server with fake responses before the legitimate response arrives, tricking the server into caching false data.

6. Infected Client Devices:

- Malware on a user's computer can change local DNS settings or modify the hosts file to redirect traffic.

Symptoms of DNS Poisoning:

1. Redirection to Fake Websites:

- Users are taken to suspicious or unrelated websites when entering known domain names (e.g., bank login pages, email portals).

2. Security Certificate Warnings:

- Browsers may display SSL certificate errors or security warnings, indicating the site does not match the certificate.

3. Unusual Network Behavior:

- Unexpected redirects, slow browsing, or communication with unknown IP addresses can occur.

4. Multiple Users Reporting the Same Issue:

- Widespread DNS redirection problems usually suggest a compromised DNS server.

5. Login Credential Theft or Malware Infection:

- Users may unknowingly submit sensitive information to attacker-controlled websites or download malicious software.

Resolving DNS Poisoning:

1. Flush DNS Cache:

- On affected machines, clear the DNS resolver cache to remove poisoned entries. ○ Command (Windows): `ipconfig /flushdns`

2. Change to Trusted DNS Providers:

- Use secure DNS services like Google DNS (8.8.8.8), Cloudflare DNS (1.1.1.1), or OpenDNS, which provide better protection and monitoring.

3. Implement DNSSEC:

- Enable DNS Security Extensions to verify the authenticity of DNS responses through digital signatures.

4. Keep DNS Software Updated:

- Regularly patch and update DNS servers to fix known vulnerabilities.

5. Monitor DNS Traffic:

- Use IDS/IPS tools and logs to detect unusual DNS activity or suspicious query responses.

6. Secure Local Networks:

- Encrypt DNS queries using DNS-over-HTTPS (DoH) or DNS-over-TLS (DoT) to prevent MITM attacks.
- Avoid using public or unsecured Wi-Fi for sensitive transactions.

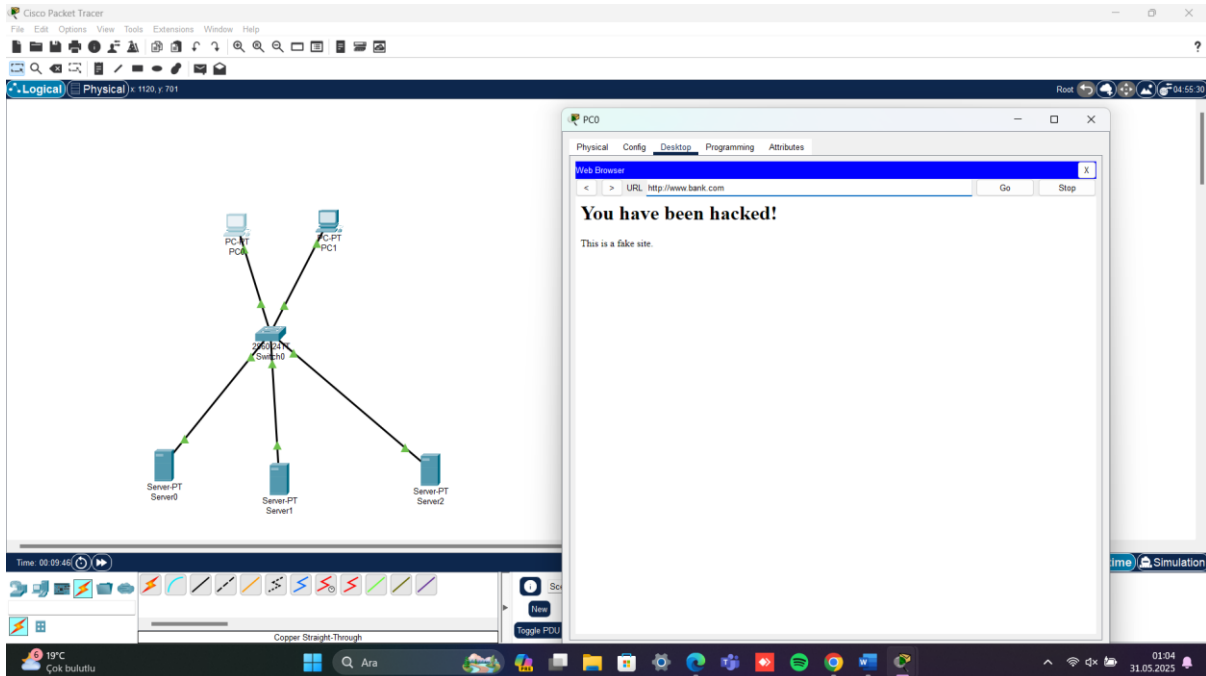
7. Educate Users on Phishing Risks:

- Training users to spot suspicious URLs, certificate warnings, and phishing pages can reduce successful DNS poisoning exploits.

8. Restrict Access to DNS Settings:

- Prevent unauthorized users or malware from modifying DNS settings on routers or endpoint devices.

Simulation of DNS spoofing : In this simulation, the IP of the given address was changed to a fake IP and when the real IP address was entered in the browser, the following image was encountered.



Simulation of solution: For the solution, the fake address (server2) was removed from the DNS server (server 0) and the real address(server1) was added instead .

